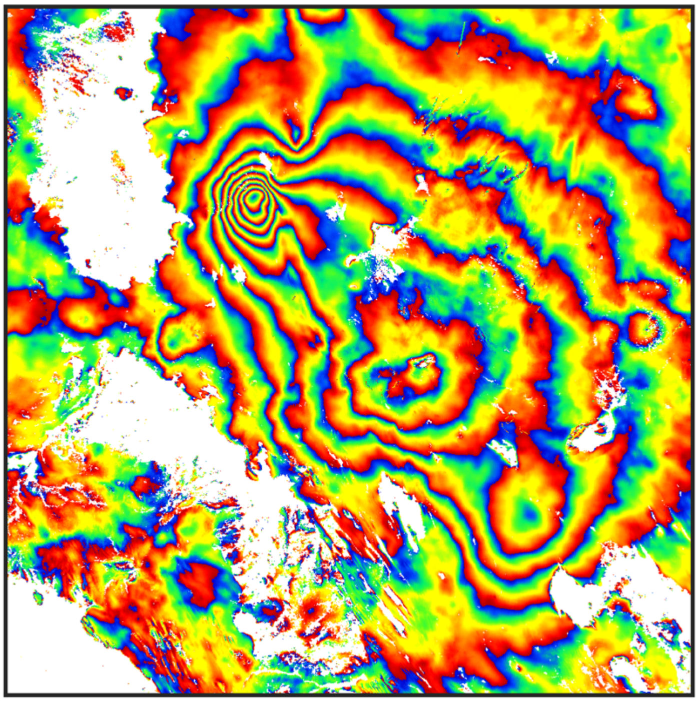
GBIS-BULK

User Manual
V1.0

An automated batch-cataloguing tool based on Geodetic Bayesian Inversion Software (GBIS)



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Overview

GBIS-BULK provides a (semi-)automated workflow in MATLAB to systematically characterise, model and cataloguing ground deformation parameters from big InSAR datasets. It is developed primarily for use in volcanic applications but could also be adapted for use in other applications e.g. earthquakes. It is built on the modular architecture of GBIS (Bagnardi and Hooper, 2018).

Batch Workflow

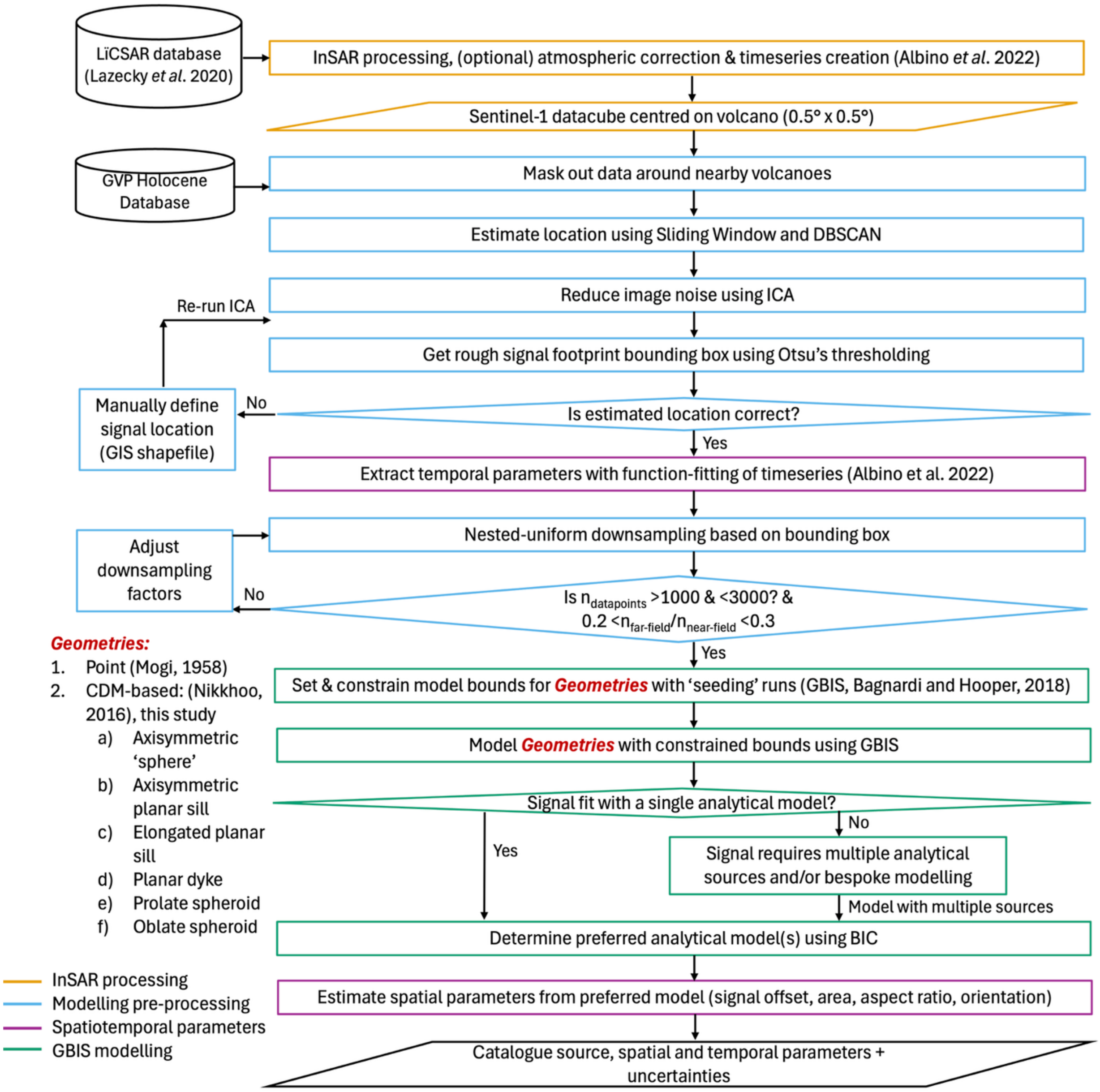
GBIS-BULK is controlled by editing and running the MATLAB script ***GBIS_BULK_Input.m***. This can be run for multiple InSAR datasets and timeseries in sequence. For computational efficiency on multi-core machines, you should run ***BatchGBISRun.sh, BatchGBISRunBG.sh, or BatchGBISRunBG_Alt.sh*** instead after configuring ***GBIS_BULK_Input.m***, which partitions different parallel jobs for each signal.

GBIS-BULK comprises of 11 main steps (step 0 to step 10), which are shown in the flowchart in Figure 1:

- **Step 0** - Convert InSAR data (timeseries or single interferogram(s)) into the correct format.
- **Step 1** - Mask out any nearby volcanoes in the scene using the Global Volcanism Programme (GVP) Holocene Volcano List *(Manual option available).*
- **Step 2** – Locate the centre of the signal automatically using clustering of areas of maximum deformation across different window sizes *(Manual option available).*
- **Step 3** – (Timeseries only) Denoise the cumulative deformation image by identifying and removing independent noise components to the signal with Spatial Independent Component Analysis (sICA) *(Manual option available)*.
- **Step 4** – Automatically extract a bounding polygon around the signal using Otsu thresholding (for use in downsampling Step 6) *(Manual option available)*.
- **Step 5** – (Timeseries only) Extract temporal parameters of the deformation signal using the function-fitting approach of Albino et al. (2022). The step uses the timeseries at the pixel identified or provided in Step 2.
- **Step 6** – Downsample InSAR data using a nested-uniform approach. This splits the image into near-field and far-field, based on the polygon calculated or provided in Step 4. In the near-field, a smaller downsampling factor is used, retaining more datapoints, and the opposite is true in the far-field area.
- **Step 7** – Prepares GBIS input file (the .inp text file) by updating model bounds to those specified in the input file and calculates the noise characteristics of the input data with an experimental semi-variogram.
- **Step 8** – Run GBIS Inversion for the input data and specified source geometries.
- **Step 9** – Post-process GBIS results and extract deformation parameters, including model spatial parameters, uncertainties and model comparison

metrics e.g. Akaike's Information Criterion (AIC). Also produces PDF reports for each model.

- **Step 10** – Catalogue all characteristics from models from one of more datasets into an updateable spreadsheet.



Below are further details on input formats and options for each step.

Input formats

Future releases hope to add compatibility for more InSAR timeseries formats e.g. MintPy. We encourage users to add compatibility for other common data formats by editing Step0 (load data) and Step3 (Remove noise with ICA).

InSAR data

GBIS-BULK supports 3 main formats of input InSAR data, either timeseries in ".nc" format, timeseries in ".h5" format (in the format of the COMET LiCSAR/LiCSBAS products), or individual interferograms in ".tif" format. Lat/Lon coordinates in all cases are assumed to be in decimal degrees. The name of the volcano should match records on the Global Volcanism Programme, if possible.

".nc" format

This format follows previous timeseries processing software and datasets such as the East African Rift dataset of Albino and Biggs (2021). A script ***Save_NC_Timeseries.m*** is provided to save a timeseries consisting of 1. Dates of all timeseries steps; 2. Lat/Lon vectors the same dimensions as the x and y of the timeseries; 3. Output filename; 4. The timeseries datacube e.g. 3D array of displacement in the Lat/Lon/Time dimensions, with displacement in metres and any incoherent areas set to zero displacement.

Output filename should be in the format "Volcano_InSARDataframe.nc" e.g. "suswa_130A_09212_131313.nc"

This should be saved in a folder named "timeseries" in a folder named "Volcano_InSARDataframe". There should also be a folder at this level called "DEM" with a geotiff DEM covering the region of interest. Consult the diagram below for full structure.

".h5" format

This format is output by timeseries generated using LiCSBAS (Morishita *et al*., 2020). The ".h5" timeseries should be saved with the rest of the LiCSBAS outputs in an enclosing folders in the format "workingDirectory/Volcano/InSARDataFrame/TS_GEOCml1clip", where "TS_GEOCml1clip" is the output directory from LiCSBAS processing.

".tiff" format

This format suits any single unwrapped InSAR interferogram in geotiff format. The displacement is assumed to be line-of-sight displacement in radians.

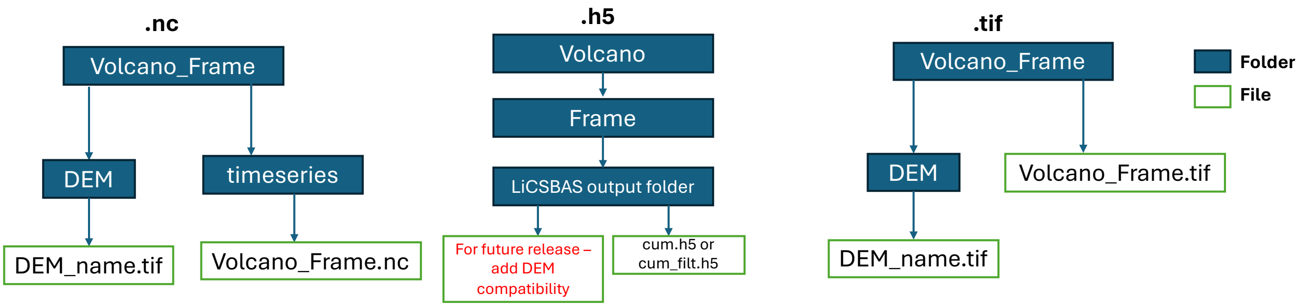


Figure 1: Recommended folder structures for different InSAR data formats

Manual Inputs

If the automatic signal location and masking in ***GBIS-BULK*** is insufficient or incorrect, you can provide manual inputs, following the folder structure below. They can be provided as a shapefile .shp (in decimal degrees coordinates) or a .mat file. **AOI** is used for a manual near-field bounding box. **Location** is used to provide a manual point for the signal centre. **Mask** is used to mask any additional areas in the image.

Full examples of the formats of these inputs can be found in the GitHub repository in the “Manual_Inputs” folder. “ID” is a unique suffix to a set of manual inputs, specified in the “MANUAL_Options.Manual_Suffix” option in ***GBIS_BULK_Input.m***.

You can specify if you have manual data to add by adjusting the manual options in ***GBIS_BULK_Input.m*** (see “Setup” and “Options” section for more details).

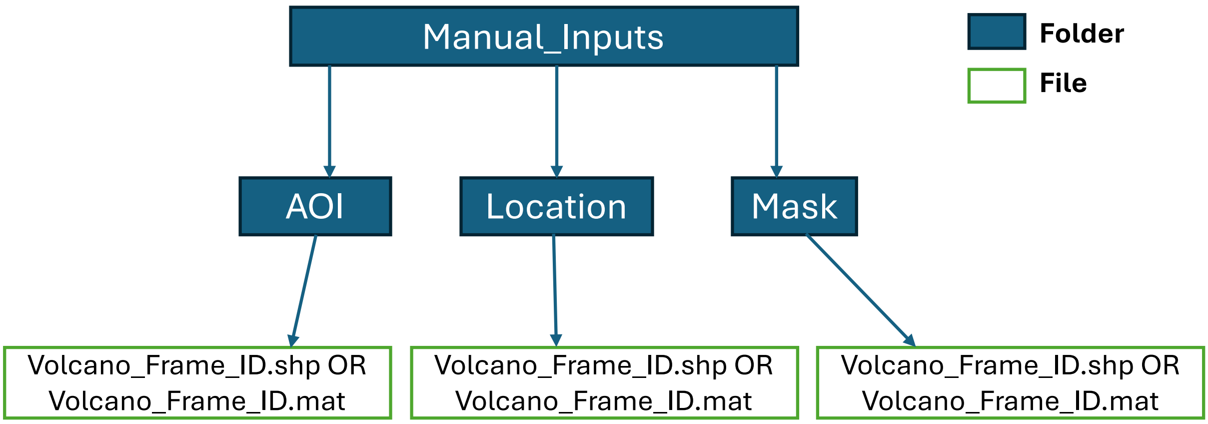


Figure 2: Folder structure for manual inputs of data into GBIS_BULK

Setup

A typical setup of ***GBIS-BULK*** comprises the following steps:

1. Create InSAR timeseries in one of the appropriate folder/format structures detailed in the “Input formats” section
2. Create any manual inputs (if desired) and add them to the folder structure
3. Edit ***GBIS_BULK_Input.m***
    a. Provide filepath to input data (with wildcards if you want to run more than 1 dataset)
    b. Provide name of .inp input file (default should be ***Generic_Input_File***) – no need to edit this unless you are manually defining model bounds
    c. Specify RunID and BulkRunID
    d. Specify start and end step (always recommended to start from step 0 to avoid unintended errors.
    e. Specify any manual inputs using the parameters starting with “MANUAL_Options”
    f. Edit parameters for each step if needed and specify model bounds (See “Options” sections below)
4. Run ***GBIS-BULK*** for the desired steps by either:
    a. Running ***GBIS_BULK_Input.m*** directly
    b. Running ***BatchGBISRun.sh*** which partitions large datasets into multiple jobs and runs them in separate sessions using tmux (requires tmux installation). Live progress is tracked in log files.
    c. Running ***BatchGBISRunBG.sh*** which partitions large datasets into multiple jobs and runs them in parallel in the background using nohup and GNU Parallel. Live Progress is tracked in log files.
    d. Running ***BatchGBISRunBG_Alt.sh*** which works the same as above but does not require GNU Parallel installation in Linux (just native bash).
    e. To stop running bulk runs at any time, find the relevant process ID or kill the relevant tmux server. Alternatively, to kill all bulk runs, run the ***stopGBIS_BatchJobs.sh*** script.

If running multiple runs or using multiple datasets, you may want to create multiple copies of ***GBIS_BULK_Input.m*** or ***BatchGBISRun.sh*** to keep track of your options.

When ran, ***GBIS-BULK*** creates a range of output folders, figures, data and reports showing the performance and outcome of each step. See the “Outputs” section for more details.

Bespoke (manual) Workflow

It is possible to run GBIS-BULK in a manual (bespoke) fashion, using some of the original GBIS scripts and updated pre/post-processing scripts.

Input formats

Currently, this can work with any input format, provided an appropriate script is written to convert the InSAR data into the correct input format, saved as a .mat file with the variables:

```
Phase – n by m array of unwrapped LOS displacement values in radians (positive away from the satellite)

Lat – n by m array of Latitude values in decimal degrees (WGS84)

Lon - n by m array of Longitude values in decimal degrees (WGS84)

Inc – n by m array of the average incidence angle across the scene in degrees

Heading – n by m matrix of the satellite heading angle in degrees

n = number of rows; m = number of columns
```

There are example scripts to convert geotiffs or LiCSBAS (Morishita et al., 2020) .h5 output files available in the folder:
GBIS_BULK/GBIS/UsefulScripts/Standalone_Scripts/

Setup

The setup steps for a bespoke manual run is:

1. Convert InSAR data into the format given above using a script in ***GBIS_BULK/GBIS/UsefulScripts/Standalone_Scripts/*** or a custom-made one if your data is not in LiCSBAS .h5 or .tif format.
2. Create an AOI to downsample around. Create a shapefile delimiting the area of interest in QGIS (WGS84 coordinates), give it a name such as VolcanoName_AOI_Name.shp.
3. Following the folder structure in the ***'Manual Inputs'*** section of this manual, copy the .shp file and auxiliary files to the ***Manual_Inputs/AOI*** folder.
4. Downsample the data by modifying and running the script at: ***GBIS_BULK/GBIS/UsefulScripts/Standalone_Scripts/SetupManualDS_Sing.m*** . You will need to specify the:
    a. Satellite wavelength in m
    b. VolcName (following the name chosen in Step 2)
    c. AOI_Name as specified in Step 2 (as MANUAL_Options.Manual_Suffix)
    d. Filepath to .mat file created in Step 1.
    e. Downsampling options (under Options.* variables) – check 'Options' section below for more information on what each one does.

    Once ran, note 1. The filepath where the downsampled data is stored; and 2. The average Sill, Range, and Nugget values outputted by the script. These specify the noise parameters of the input data.

If you have multiple tracks of data e.g. ascending and descending, repeat steps 1-4 for each dataset. Note: for each dataset, a new copy of the .shp file from Step 2 with a new name needs to be made.

5. Make your .inp input file. Make a copy of ***Generic_Input_File.inp*** and change the following rows:
   a. geo.referencePoint – Change to the bottom left (minimum) Lat and Lon values from the .mat file created in Step 1.
   b. geo.boundingBox – Specify the bounding Lat and Lon coordinates of the InSAR data, using the .mat file created in Step 1.
   c. For each set of InSAR data:
      i. insar{insarID}.dataPath– path to downsampled data created in Step 4
      ii. insar{insarID}.rawDataPath– path to full resolution data created in Step 4
      iii. insar{insarID}.constOffset – specific constant offset or not
      iv. insar{insarID}.rampFlag – specifiy remove linear ramp or not
      v. insar{insarID}.sillExp, insar{insarID}.range, insar{insarID}.nugget – specify noise parameters output in Step 4.

   Note: Leave the model-specific parameters in the .inp file for now, we will modify them later.

6. Setup your script to run GBIS manually, by making a copy of ***GBIS_BULK/GBIS/UsefulScripts/Standalone_Scripts/RunGBIS_Manual.m*** and editing the script to point to your .inp file created in Step 5, specify the InSAR data you want to use and the models you want to run. Also modify the output filenames at the bottom of the script. The script will iteratively run GBIS for all models specified, then create output figures and tables for them.

7. To start, run the script created in Step 6. It will likely throw an error when starting the inversion. If not, press ctrl+C to stop the script once the inversion begins. This is so GBIS generates figures of our data in its local coordinate system, so you know which model bounds to search. These figures can be found in:
   GBIS_BULK/Inversion_Results/Name_Of_Inp_File/invert_*/invert_*/Figures/

8. Open the .inp file created in Step 5, and modify the model-specific input parameters to the ranges you want to explore, and for the models you want to test.

9. Once you have updated the .inp file, you can re-run the script created in Step 6. Depending on how many datapoints and models you have, and your hardware, it could take <30 minutes to many hours in run through. Please see the 'Outputs' section of this manual to understanding the output figures and tables produced by GBIS-BULK.

10. [Optional] You can modify the script at:
   GBIS_BULK/GBIS/UsefulScripts/Standalone_Scripts/MakeOutputFigsTa

bles.m to point to your inversion results, which can be used to re-create the tables and create nicer data-model-residual figures.

Options

Below are the options for ***GBIS-BULK*** listed in ***GBIS_BULK_Input.m*** and their description and dependencies. Note that any parameters with a strikethrough do not currently work and should not be changed and will be updated or removed in a future release. The majority of pre-processing options should be kept at default values unless there are good reasons to change them.

Required Parameters

Input_Filename – Name or filepath of .inp file to edit (default – Generic_Input_File.inp)

TS_Files – dir function call to identify all InSAR timeseries files to use in the bulk inversion e.g. dir([pwd,'/SampleData','/**/timeseries/*.nc'])

General Parameters

SaveOptions-Save parameters

~~**LoadOptions**~~

SaveProgress – Save progress – useful if testing a few steps at once

~~**LoadProgress**~~

~~**LoadFilename**~~

Options.SpatialRes-Approximate spatial resolution of the input InSAR data in m – used in calculating signal area and offset.

Options.WavelengthM – Wavelength of the SAR sensor in m e.g. 0.056 for Sentinel-1

Options.RunID – Unique ID for each run

Options.BulkRunID – Unique ID for each set of bulk runs

Options.StartStep – Start step of GBIS-BULK (recommended 0)

Options.EndStep – End step of GBIS-BULK (recommended 10)

~~**Options.SkipGBIS**~~

~~**Options.SkipFlagged**~~

Manual options

MANUAL_Options.Manual_Suffix – Unique suffix for all manual input filenames relating to a single manual run (see file structure)

MANUAL_Options.MaskData – Manually mask regions of the image (1) (Aside from or along with masking out nearby volcanoes) or not (0)

MANUAL_Options.MaskDataFmt – Data format of masking data ('shp' – shapefile; 'mat' – Matlab file)

MANUAL_Options.RegionShrink – Optionally apply region shrinking to the input data manually with a disk element (parameters below) (1==yes; 0==no)

MANUAL_Options.RegionShrinkMinPts – Minimum size of region (in pixels) that region shrinking is applied to.

MANUAL_Options.RS_DiskSize – Distance to shrink region by (in pixels)

MANUAL_Options.AOI – Alternative for Step 2 and Step 4 (1==yes; 0==no)

MANUAL_Options.AOIFmt – Data format of manual AOI data ('shp' – shapefile; 'mat' – Matlab file)

MANUAL_Options.Location – Alternative for Step 2, Step 4 and Step 5 – need to also have MANUAL_Options.AOI == 1 to use

MANUAL_Options.LocationFmt – Data format of manual Location data ('shp' – shapefile; 'mat' – Matlab file)

MANUAL_Options.ICA – Alternative for Step 3 – do ICA with manually inputs (below) (1==yes; 0==no)

MANUAL_Options.ICA_nComp – Number of components e.g 5

MANUAL_Options.ICA_White – Optionally whiten input data (RECOMMENDED) (1==yes; 0==no)

MANUAL_Options.ICA_AllComps – Take all IC components for reconstructed data (1==yes; 0==no)

MANUAL_Options.ICA_ChosenComps – Take selected IC components for reconstructed data e.g. [1, 2, 4] for IC 1, 2 and 4.

Step 0 – Initial pre-processing of InSAR data

Options.AutoAngles – Automatically retrieve head and inc angles from LiCSAR (requires LiCS frames) – otherwise provide them manually in cmd line (1==yes; 0==no)

Options.CropTS – Crop timeseries? (1==yes; 0==no)

Options.CropTS_startAll – Start timestep to crop timeseries (if Options.CropTS ==1). 1 value per frame e.g. [3 2] for two frames.

Options.CropTS_endAll – End timestep to crop timeseries (if Options.CropTS ==1). 1 value per frame e.g. [29 32] for two frames.

Options.CropImg – Spatially crop image? (1==yes; 0==no)

Options.CropImgX – X range to crop image (if Options.CropImg ==1) e.g. if image is 500x500, 100:200 will retain 100 pixels in X direction

Options.CropImgY – Y range to crop image (if Options.CropImg ==1) e.g. if image is 500x500, 100:200 will retain 100 pixels in Y direction

Options.IgnoreLastStep – Optionally remove first and last timeseries step (can reduce noise in some cases) (1==yes; 0==no) (if Options.AutoTimestep ==0)

Options.AutoTimestep – Optionally choose between the last and second-last step of the timeseries, choosing the one with lowest stdev (1==yes; 0==no)

Step 1 – Masking nearby GVP Volcanoes

Options.MaskVolcs – 1= Mask x km radius around all other volcanoes in the ifg that aren't the target. 0= don't mask

Options.Mask_BufferDist – Radius to mask around volcanoes in pixels (distance = n.pixels * spatial resolution) – distance is approximate

Step 2 – Estimate signal centre using sliding window approach

~~Options.SlidingWindow~~

Options.SW_WindowSizes – Range of window sizes to compared e.g. Default =1:25, compares 1x1,2x2 etc… to 25x25.

Options.SW_CropEdges - Crop outer X% of the images? (1==yes; 0==no)

Options.SW_CropEdges_Mag - Portion of the image edge to crop (if Options.SW_CropEdges==1) e.g. Default = 0.2 = 20%

Options.SW_RegionShrink - Apply region shrinking to remove edges of NaN areas (poor unwrapping) (1==yes; 0==no)

Options.SW_RegionShrink_MinPts - Minimum size of region (in pixels) that region shrinking is applied to (if Options.SW_RegionShrink ==1) (Default = 500)

Options.SW_RegionShrink_DiskSize - Distance to shrink region by (in pixels) (if Options.SW_RegionShrink ==1)(Default = 3)

Options.SW_NaN_Thresh - Minimum proportion of non-Nan pixels in a sliding window region for mean value to be retained (Default = 0.95)

Options.SW_DBSCAN_Eps - Epsilon parameter for DBSCAN (Default = 50)

Options.SW_DBSCAN_MinPts - MinPts parameter for DBSCAN (Default = 7)

Options.SW_MinBBSize - Minimum size of initial signal location region i.e. x by x pixel square (Default = 20)

Step 3 – Noise reduction using ICA

Options.ICA - 1=Use ICA to de-noise timeseries prior to Otsu; 0=Don't use ICA (use original data)

Options.ICA_nCompRange - Range of ICs to run ICA with (repeats can be used to account for impacts of random start position) - (Default = [5 5 5 10 10 10 20 20 20])

Options.ICA_ClosenessThresh - What proportion of the signal can be considered noise and removed by ICA e.g. 0.95 means 5% of the signal magnitude can be discarded as noise (Default = 0.95)

Step 4 – De-limit near-field region using Otsu Thresholding

Options.Otsu_NumLevels – Number of levels of Otsu thresholding to test (number of segments = numlevels + 1) (Default = 5)

Options.Otsu_BB_Shape – Make Otsu region rectangular (1) or the shape of the region (2, recommended) (Default = 2)

Options.Otsu_RegionShrinkCohThresh – Don't perform region shrinking if non-zero/NaN pixels make up less than X proportion of the image. (Default = 0.25)

Options.Otsu_RegionShrink_MinPts – Minimum size of region (in pixels) that region shrinking is applied to (if Options.SW_RegionShrink ==1) (Default = 500)

Options.Otsu_RegionShrink_DiskSize – Distance to shrink region by (in pixels) (if Options.SW_RegionShrink ==1)(Default = 3)

Options.Otsu_MinClassSize – Minimum size (in pixels) of an Otsu class to be retained (Default = 200)

Options.Otsu_MaxOverlap – Minimum overlap between the signal location from Step 2 (Location.Limits) and an Otsu class needed for the Otsu class to be considered as defining the near-field. (Default = 0.8)

Options.Otsu_MinConCompSize – Minimum size (in pixels) of connected component region in the chosen class to be retained. (Default = 50)

Options.Otsu_RegionConnectivity – Region connectivity criteria for above (4= connected by edges), (8= connected by edges or corners). (Default = 8)

Options.Otsu_SizeLimit – What proportion of the whole image the Otsu region (FineBoundingBox) can contain (Default = 0.75)

Options.Otsu_MaxMaskArea – Threshold max region area (in pixels) below which buffering is applied (Default = 5000)

Options.Otsu_Buffer_Large – Number of pixels to buffer large Otsu regions (bigger than Options.Otsu_MaxMaskArea pixels) (Default = 1)

Step 5 – Modelling timeseries by fitting function

Options.Temp_R2_Thresh – R2 threshold for function fits above which the function can be considered (Default = 0.5)

Step 6 – Nested-uniform downsampling of InSAR data

Options.SS_Factor – Starting Coarse downsampling factor i.e. anywhere outside the AOI is averaged over every x by x pixels (Default = 20)

Options.SS_FactorF – Starting Fine downsampling factor i.e. anywhere outside the AOI is averaged over every x by x pixels (Default = 4)

Options.NaN_Thresh_DS – Proportion of NaN values in an x by x averaged block above which the downsampled pixel is given a NaN value (Default = 0.5)

Options.Adjust_SS_Factor – Automatically adjust the SS_Factors based on the number of pixels if they are outside the ranges below. (1==yes; 0==no)

Options.SS_RunLimit – Limit of number of times SS_Factor can be adjusted before taking the result (if Options.Adjust_SS_Factor == 1) (Default = 20)

Options.Min_nPix – Minimum number of pixels for a downsampled imaged (if Options.Adjust_SS_Factor == 1) (Default = 1000)

Options.Max_nPix – Maximum number of pixels for a downsampled imaged (if Options.Adjust_SS_Factor == 1) (Default = 3000)

Options.Max_SS_Factor – Maximum subsampling factor in far-field (Default = 30)

Options.Max_SS_FactorF – Maximum subsampling factor in near-field (Default = 10)

Options.MinPropFF – Minimum proportion of subsampled far-field points relative to near-field points (Default = 0.2)

Options.MaxPropFF – Maximum proportion of subsampled far-field points relative to near-field points (Default = 0.3)

Options.FarFieldMask – Optionally mask additional far-field pixels (1 == yes; 0 ==no)

Options.FarFieldUnmaskedVar - Use unmasked far-field image for semi-variogram generation rather than masked far-field (1 == yes; 0 ==no) (If Options.FarFieldMask == 1 AND Options.Variogram == 1)

Options.FarFieldMaskMethod - Mask far-field pixels based on 1. Areas where DEM is +/- N std away from near-field mean OR 2. Additional buffer of near-field region (if Options.FarFieldMask == 1) OR 3. combine both methods. (Default = 2)

Options.FarFieldDEM_StdLimit - N of std away from near-field mean elevation to mask data (if Options.FarFieldMaskMethod == 1 OR 3) (Default = 1.5)

Options.FarFieldAdditionalBuffer - Percentage of original image size to keep outside of near-field region (if Options.FarFieldMaskMethod == 2 OR 3) (Default = 15%)

Options.PreProcOffset - Minimise far-field signal by removing offset during preprocessing? E.g. NewImg = Img - mean_of_far-field_pixels, (1==on, 0==off) (Default = 1)

Step 7 part 1 – Prepare model bounds and input file

Options.EstimateDepthBounds - Optionally estimate depth bounds based on size of Otsu region and Mogi equation (depth vs signal size) (1==yes, 0==no). Relies on a precise Otsu region to be accurate. (Default = 0)

Options.DepthEstimateA - Estimate of proportion of maximum LOS displacement at the edge of the near-field bounding box (if Options.EstimateDepthBounds==1) (Higher = deeper depth constraints) (Default = 0.05)

Options.Offset - Invert for constant offset in GBIS? (1==yes, 0==no) (Default = 1)

Options.OffsetStep - Step bounds for constant offset (if Options.Offset ==1) (Default = 1e-4)

Options.OffsetLower - Lower bounds for constant offset (if Options.Offset ==1) (Default = -0.1)

Options.OffsetUpper - Upper bounds for constant offset (if Options.Offset ==1) (Default = 0.1)

Options.Ramp – Invert for linear ramp in GBIS? (1==yes, 0==no) (Default = 0)

Options.RampStep – Step bounds for linear ramp (if Options.Ramp ==1) (Default = 1e-7)

Options.RampLower – Lower bounds for linear ramp (if Options.Ramp ==1) (Default = -2e-6). Maximum displacement from ramp = Options.RampUpper * max local X/Y coordinate (in m) E.g. +/- 2e-6 m at 50 km grid = ramp with +/-0.1 m across the whole image.

Options.RampUpper – Upper bounds for linear ramp (if Options.Ramp ==1) (Default = 2e-6). Maximum displacement from ramp = Options.RampUpper * max local X/Y coordinate (in m) E.g. +/- 2e-6 m at 50 km grid = ramp with +/-0.1 m across the whole image.

Options.Variogram - Specify if to calculate noise parmaeters from semi-variogram; 0 = default values, 1 = use variogram, other = use hard-coded values in input file. (Default = 1)

Options.VariogramAttempts - (if Options.Variogram==1) Number of variogram runs to use (range, sill and nugget values taken as averages of these runs) (Default = 20)

Options.Bounds - Modify model bounds or not (in the .inp file). Useful when manually setting up bounds in input files. (Default = 1)

Options.BoundLimits - Modify location bounds (XY) based on 1. Geometry of the fine bounding box, or 2. Geometry of the image i.e. explore the full image (Default = 1)

Options.Change_Wavelength - Modify wavelength or not (default: 0.056 m). If modifying, change this to the desired wavelength value in m. (Default = 0)

Step 7 part 2 – Source geometries

Options.SourceType - Default initial source types - 1=Mogi (Mogi, 1958) 2=Penny-shaped crack (Fialko et al., 2001) - other sources, see below (change in Step8_RunBulkInversion script) (Default = 1)

Options.PennyComparison - Do a comparison with a Penny source (3 geometry options) and compare with other sources (1 = yes; 0 = no)

Options.SillComparison – Do a comparison with an Okada Sill source (Okada, 1985) and compare with other sources (1 = yes; 0 = no)

Options.McTigueComparison – Do a comparison with a Spherical McTigue source (McTigue, 1987) and compare with other sources (1 = yes; 0 = no)

Options.YangComparison – Do a comparison with a Yang (Spheroid) source (3 geometry options) and compare with other sources (1 = yes; 0 = no)

Options.DykeComparison – Do a comparison with an Okada Dyke source (Okada, 1985) and compare with other sources (1 = yes; 0 = no)

Options.CDMComparison – Do a comparison compound dislocation model (CDM) source Nikkhoo (2017) (Various sub-geometries – see Options.CDMGeometry)

Options.PennyType – Use (Fialko et al., 2001) (1), Sun (1969) Penny-shaped crack solution using pressure (2), or Sun (1969) Penny-shaped crack solution using volume (3)

Options.YangType – Yang Type (1= Yang (1988); 2=Cervelli (2013) spheroid (volume); 3=Cervelli (2013) spheroid (pressure))

Options.CDMGeometry – Constrain CDM to particular geometry or geometries (see below) – single number for one geometry, multiple for more than 1 geometry e.g. [1,3,6]:

- 1 = CDM with full freedom (all params)
- 2 = simple axisymmetic sphere (x,y,z,r,dV)
- 3 = axisymetric sphere (with trend/plunge) (x,y,z,r,tr,pl,dV)
- 4 = sill (symmetric axes) (horizontally planar) (x,y,z,a,dV)
- 5 = sill (non-symmetric axes) (x,y,z,r,a,b,dV)
- 6 = dyke-like (vertically planar) (x,y,z,l,w,str,dV)
- 7 = prolate spheroid (x,y,z,a,ar,tr,pl,dV)
- 8 = oblate spheroid (x,y,z,a,ar,tr,pl,dV)

Options.OtherSource – Optionally try to model with another source or combination of sources other than those above (0== No, 1== Yes)

Options.OtherSourceType – Additional source type if Options.OtherSource == 1 (You can add multiple sources e.g. 'MD' but bounds have to be set manually)

Step 8 – GBIS model setup

Options.nRuns – Number of iterations for GBIS for main inversion (Default = 200,000)

Options.SeedingRun – Run seedings runs to constrain model bounds further from their initial wide 'open' state. (1 = yes; 0 = no) (Default = yes)

Options.SeedingMaxRuns – Number of seeding runs to run in sequence to constrain bounds. (Default = 3)

Options.SeedingnRuns – Number of iterations for a seeding run (if Options.SeedingRun ==1) (Default = 50,000)

Options.SeedingBurnin – Burn-in (nRuns) for seeding runs for setting lower and upper bounds from percentiles. (Default = 30,000)

Options.SeedingMethod – Method for constraining bounds (1== Percentiles of seeding run results; 2== stdDev of seeding run results) (Default = 2)

Options.SeedingCriteria – Criteria for new bounds. If Options.SeedingMethod == 1, Options.SeedingCriteria == percentiles e.g. 95; If Options.SeedingMethod == 2, Options.SeedingCriteria == N stdDevs from optimal results. (Default = 2)

Options.SeedingTargetReduction – Target mean reduction (%) in bound ranges across all parameters (to allow seeding runs to exit early). If you don't want to set a target, set it to 100%) (Default = 75%)

Options.skipSimulatedAnnealing – Skip simulated annealing process in GBIS. (Default = 'n' (no))

Options.SingleFrameOnly – Only run 1 frame per volcano (1==yes;0==no) (Default = 0)

Options.AscDscOnly – Only run for InSAR data you have more (1==yes;0==no) than 1 frame of data for (Default = 0).

Step 9 – Create reports

Options.Reports – Generate final reports at the same time or not (1==yes;0==no) (Default = 1)

Options.AreaCutoff – (in m) – cutoff value to use when calculating area of signals from forward models – this should relate to the noise of the input images (Default = 0.012 m)

Options.Burnin – Burn-in value (number of runs) used when preparing the final reports for the main run (if reports are used) (Default = 50,000)

~~**Options.BulkOutputs**~~

~~**Options.BulkOutputsRaw**~~

~~**Options.BulkUnwrapped**~~

~~**Options.BulkWrapped**~~

Options.CreateTable – Create output table or not (1==yes;0==no) (Default = 1)

Step 10 – Create deformation catalogue

Options.TableOverwrite – Overwrite table if filename is the same (1==yes;0==no) (Default = 0)

Options.TableRoundValues – Round numeric values in table (0==no; >0 == number of sig. figs. to round to) (Default = 5)

Outputs

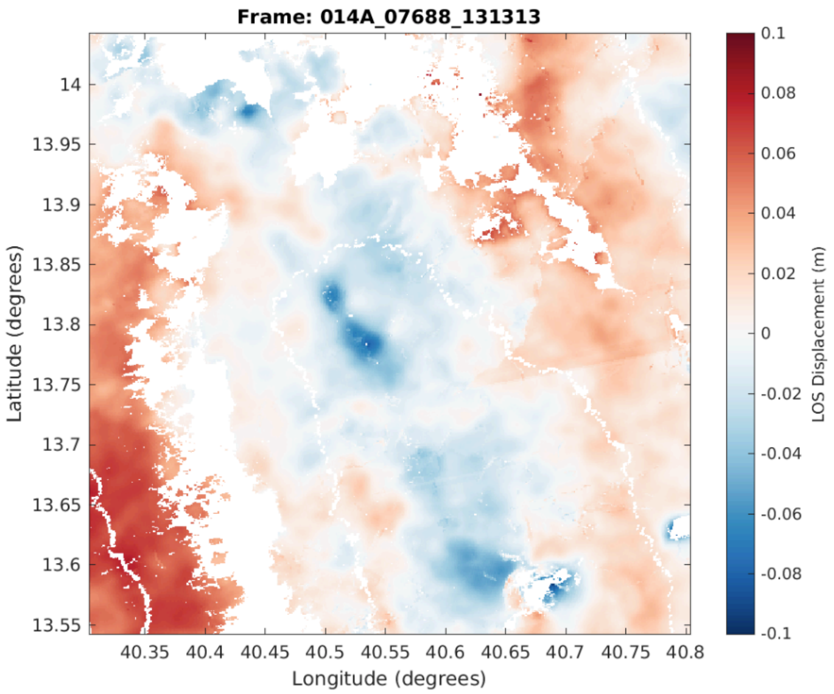
When ran, ***GBIS-BULK*** creates a range of output folders, figures, data and reports showing the performance and outcome of each step. The subsections below show the main outputs from each step, what they contain and where they are saved.

Step 0 – Initial pre-processing of InSAR data

Outputs saved in “Original Data” folder:

- Volcano_Frame.png – Cumulative LOS displacement map (example below)

- Volcano_Frame.mat – Cumulative LOS displacement map + lat, lon, heading and incidence angles



Outputs saved in "LiCSMetadata" folder:

- Metadata for the LiCSAR frame (if using LiCSAR Sentinel-1 data)

Outputs saved in "Options" folder:

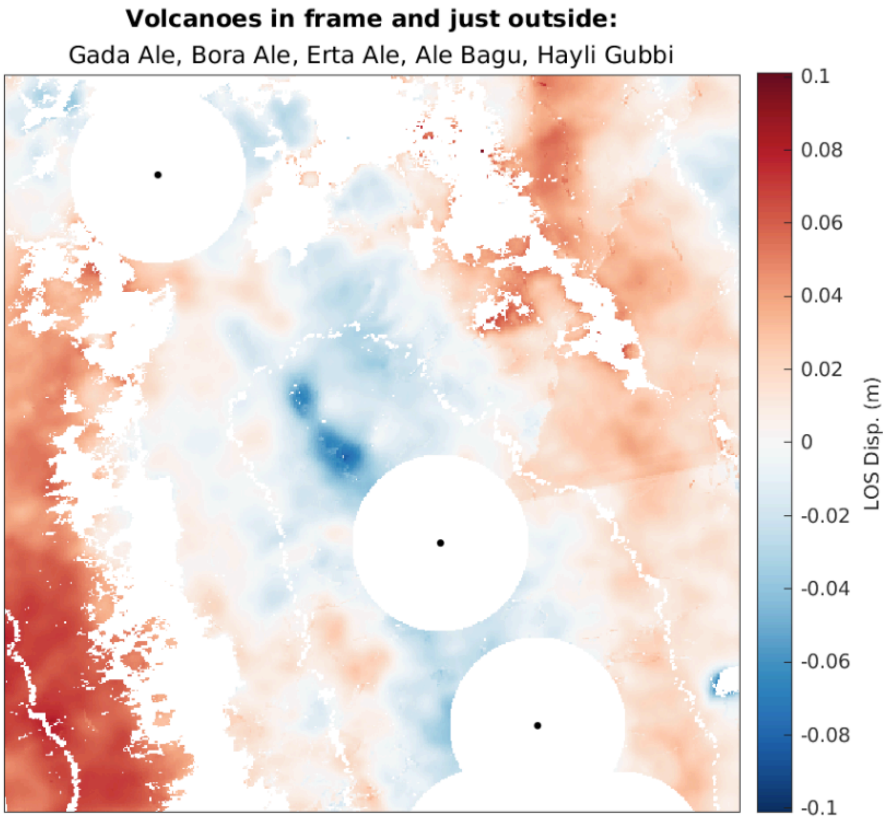
- BulkID_RunID_Options.mat – values for all options for that run

Step 1 – Masking nearby GVP Volcanoes

Outputs saved in "Buffers" folder:

- Volcano_Frame_Buffer.png – Image showing areas masked through buffering (example below)

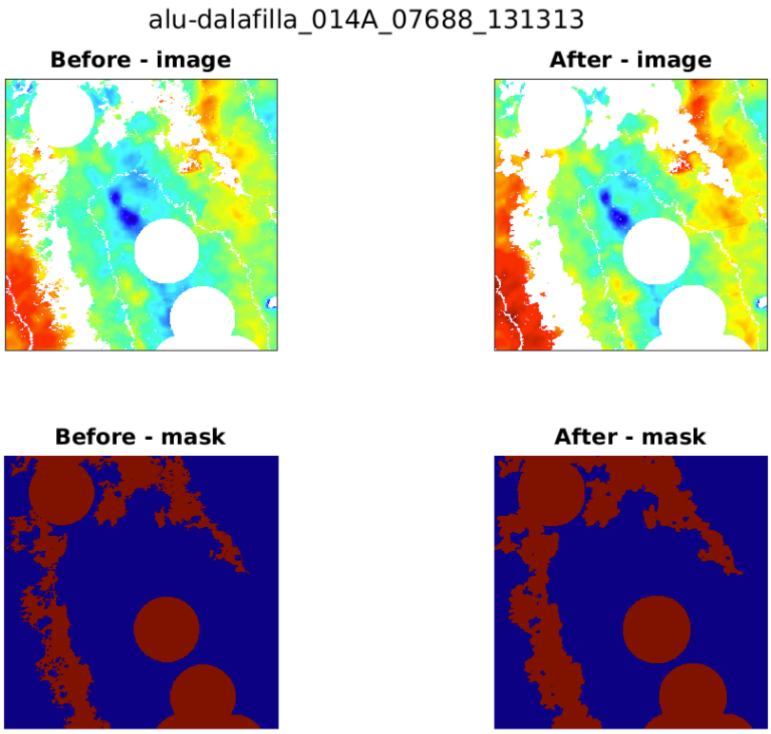
- Volcano_Frame_Buffer.mat – Cumulative LOS displacement maps before and after buffering

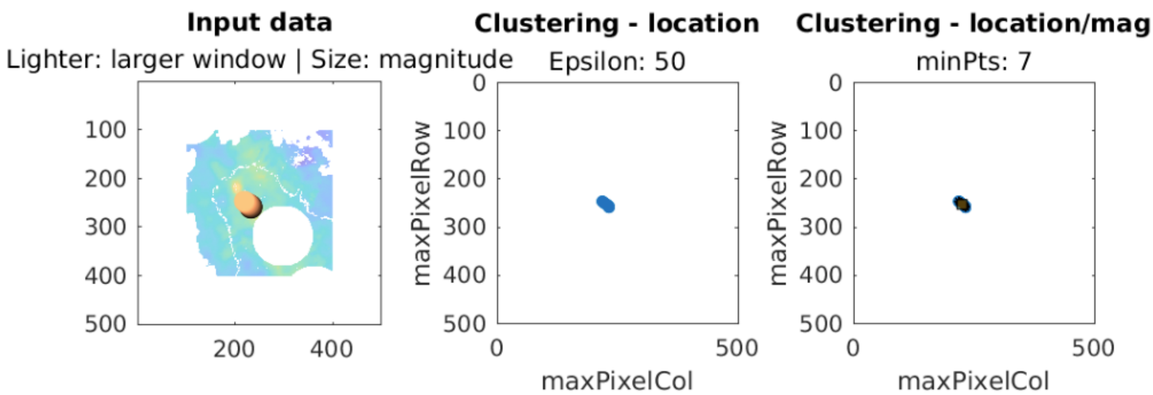


Step 2 – Estimate signal centre using sliding window approach

Outputs saved in “SignalLocation” folder: (* denotes any pattern)

- Volcano_Frame _RegionShrink.png – Figure showing results of region shrinking procedure (example below)
- Volcano_Frame_SlidingWindow_Clustering*.png – Figure showing results of the sliding window and clustering approach (example below)
- Volcano_Frame_SlidingWindow_Clustering*clean.png – Uncluttered version of the above figure
- Volcano_Frame_SlidingWindow_Clustering*.mat – Stats and results of the sliding window and clustering approach.

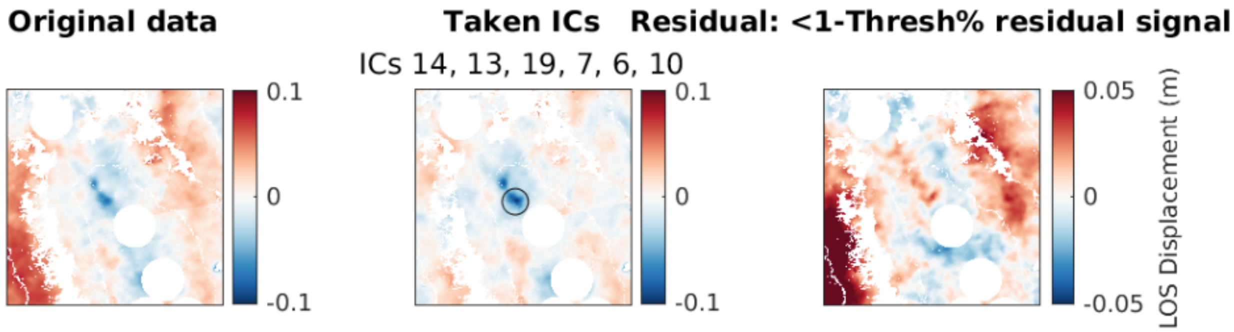




Step 3 – Noise reduction using ICA

Outputs saved in “ICA” folder:

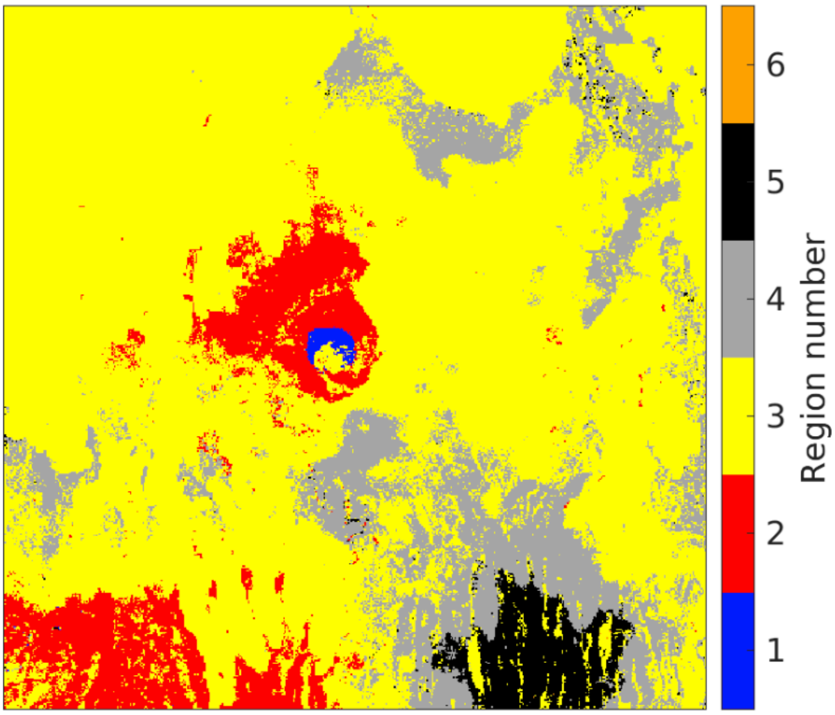
- ICA_Volcano_Frame_RUNIDComparison.png – Image showing the original data and denoised data from ICA with the residual between the two (example below).
- ICA_Volcano_Frame_RUNIDComparison.mat – Statistics and data from ICA including 1. Displacement maps and timeseries for original data and each independent component; 2. Stats relating to reach iteration of ICA; 3. RMS reduction from the ICA; 4. Note on when and if the ICA procedure was successful in denoising the data.



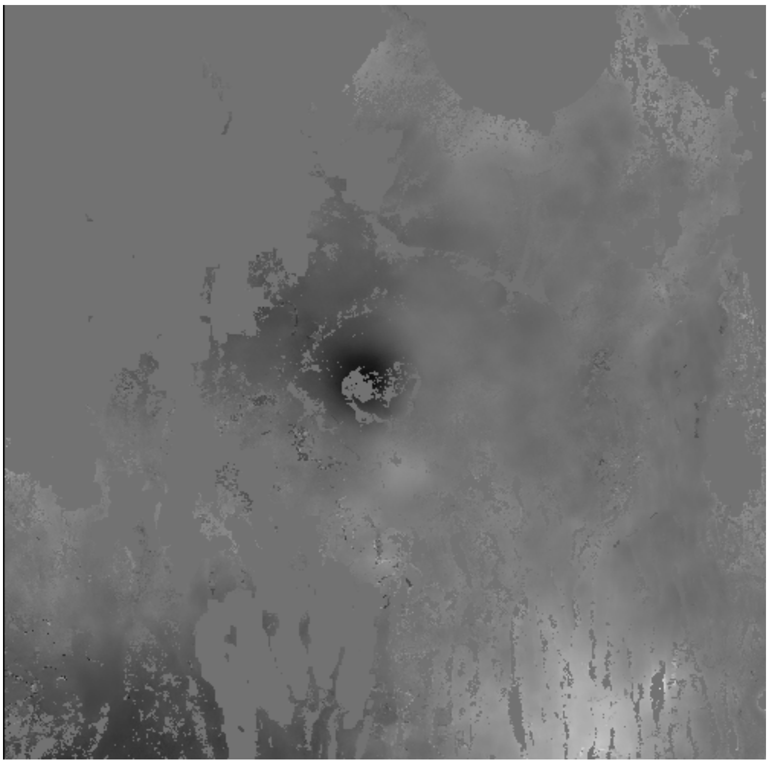
Step 4 – De-limit near-field region using Otsu Thresholding

Outputs saved in “Bounding_Boxes” folder:

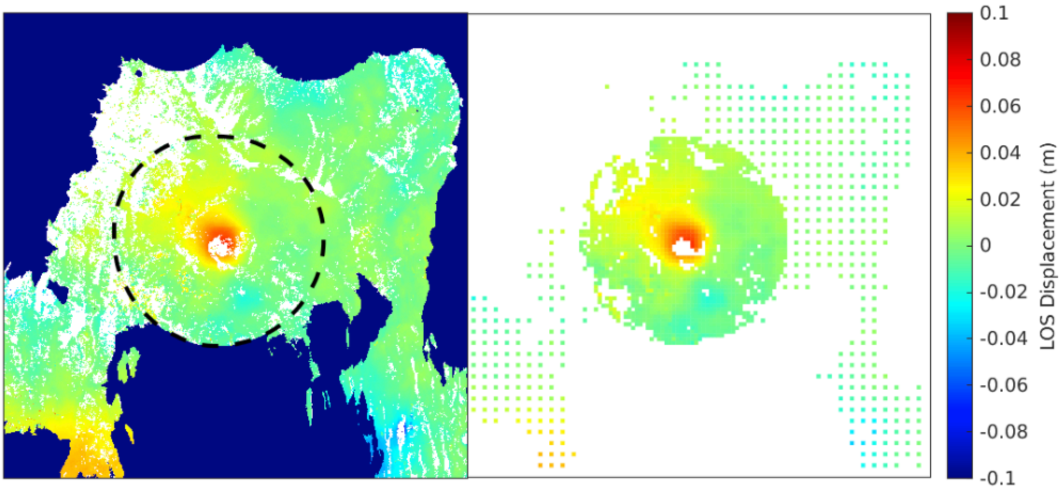
- Volcano_Frame_BoundingBox_*_RUNID.mat – MATLAB Polyshape in pixel coordinates delimiting the near-field area.
- Volcano_Frame_LargestMask_*_RUNID.mat – N x M mask of the chosen Otsu component used to delimit the near-field and far-field area, prior to buffering, where N x M is the size of the deformation image.
- Volcano_Frame_RegionShrinkResults_*_RUNID.mat – N x M matrix of the deformation image before and after region shrinking.
- Volcano_Frame_Segments_*_RUNID.mat – N x M x L Otsu segmentation results, each datapoint represents a group number, for each of the L segmentations performed. (Default L = 5)
- Volcano_Frame _BoundingBox.mat – Contains:
    - BoundingBox and BoundingBoxPoly - MATLAB polyshape and vector of max and min coordinates of the deformation image
    - FineBoundingBox and FineBoundingBoxLL – MATLAB polyshape of the near-field area in pixel and lat/lon coordinates
    - Largest_component_mask – *same output as “Volcano_Frame_LargestMask_*_RUNID.mat”*
- Volcano_Frame_Segment_*_*_RUNID.png – Otsu segmentation of the deformation image for each number of segments tested.



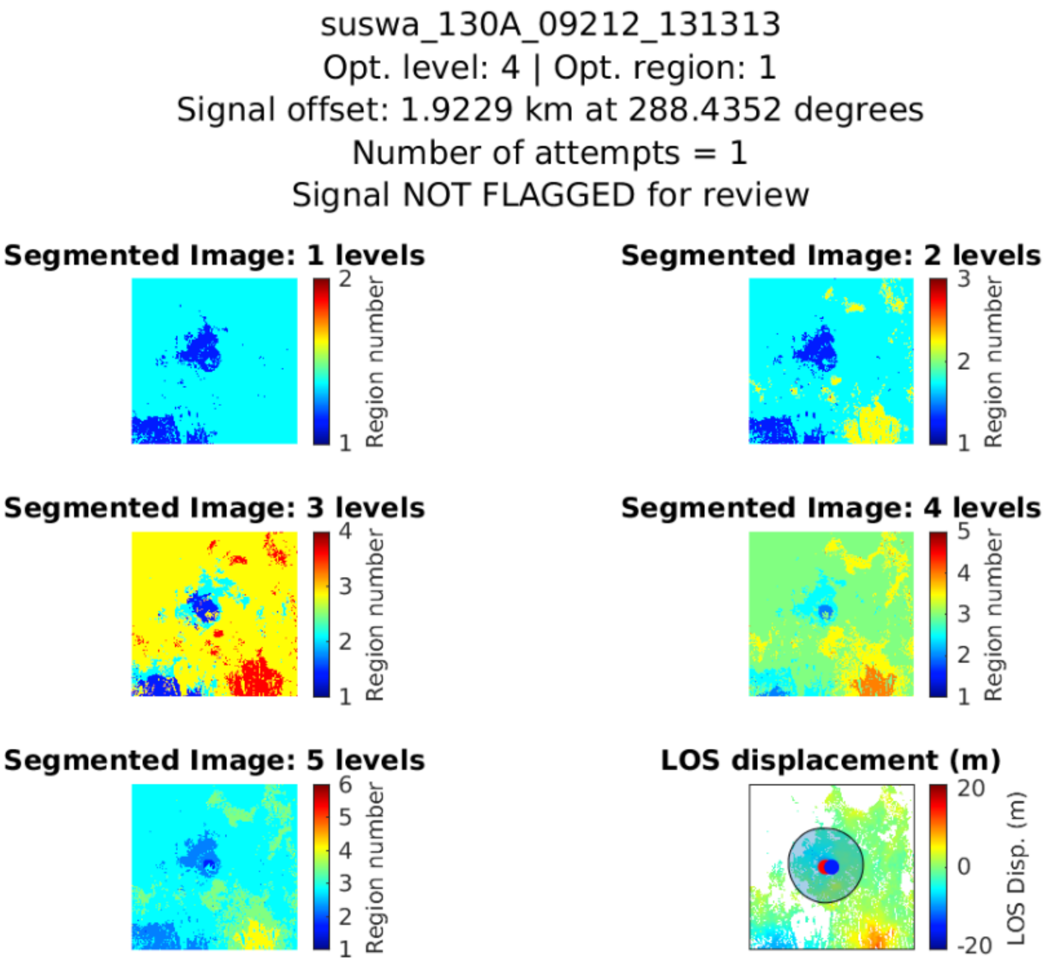
- Volcano_Frame_GrayIfg_Shape_*_RUNID.png – Grayscale version of the deformation image used in Otsu thresholding



- Volcano_Frame_OtsuDSFig_Shape_*_RUNID.png – Before/After image of downsampling – blue areas in the left panel show additional far-field areas removed based on a DEM.



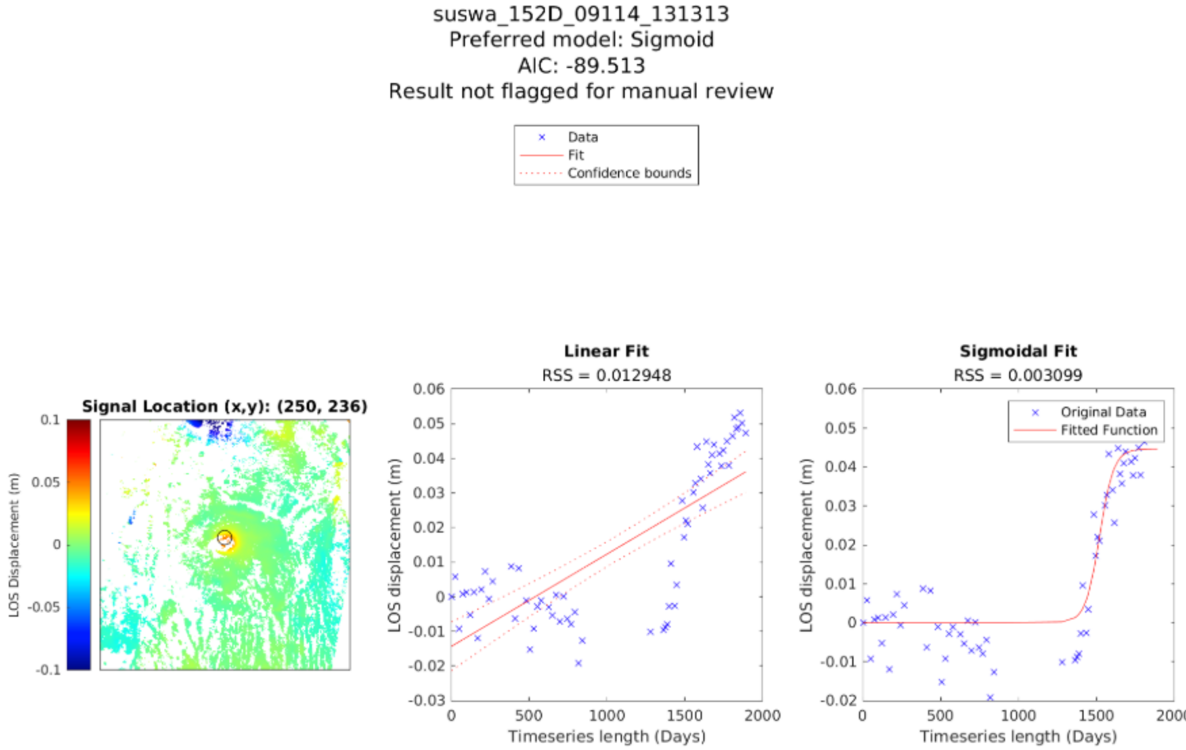
- Volcano_Frame_OtsuFig_Shape_*_RUNID.png – Overview of Otsu thresholding results and near-field/far-field area



Step 5 – Modelling timeseries by fitting function

Outputs saved in the “Temporal_Parameters” folder:

- TempPara_Volcano_Frame.mat – Contains "Model" structure giving details of the chosen model, ΔAIC value and associated temporal parameters.
- TempPara_Volcano_Frame.png – Gives an overview of the temporal fits of linear and sigmoidal models and which is preferred.



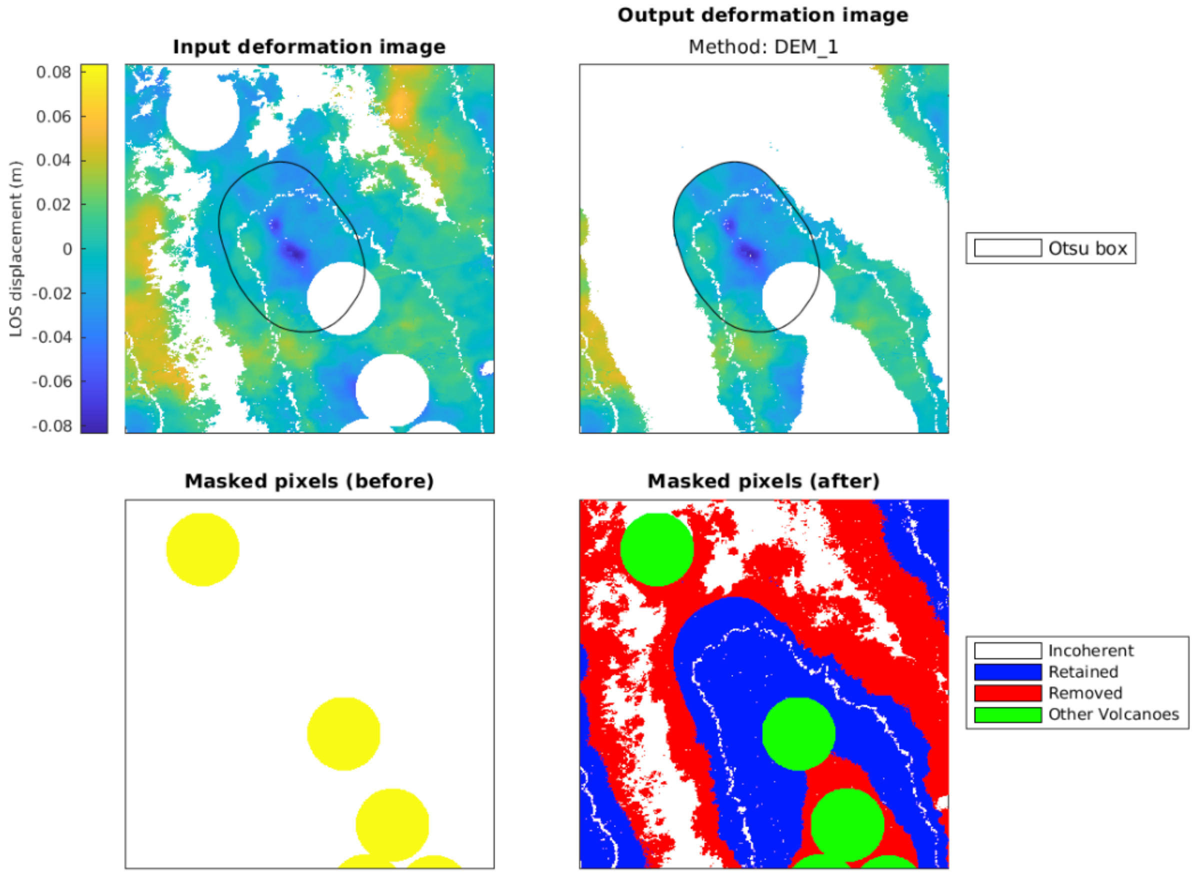
Step 6 – Nested-uniform downsampling of InSAR data

Outputs saved in the "InputData" folder:

- Volcano_Frame_DownsamplingStyle_RUNID.mat – Contains:
  - Vectors of Lon/Lat/Phase/Heading/Incidence in GBIS format
  - Offset – details of offset removed if *Options.PreProcOffset=1*.
  - FineBoundingBox – Polyshape of near-field extent
  - Coarse/FineIdxs – Vector indices of downsampled pixels that are in the near-field and far-field areas.
  - DS_Stats – Statistics of each downsampling iteration showing downsampling factors used and number and proportion of near-field and far-field datapoints.
  - nObs_Raw/SS/Coarse/Fine – Number of raw datapoints, downsampled datapoints, far-field datapoints and near-field datapoints
- Volcano_Frame_No_SS_RUNID.mat
  - Vectors of Lon/Lat/Phase/Heading/Incidence in GBIS format
  - Offset – details of offset removed if *Options.PreProcOffset=1*.
  - FineBoundingBox – Polyshape of near-field extent
  - FullResLat/Lon/Phase – Arrays of Lat/Lon/Phase

Outputs saved in the “FarfieldMasks” folder:

- DEM_Volcano_Frame.png – Image showing far-field pixels removed from the deformation image based on the DEM if relevant options are selected.



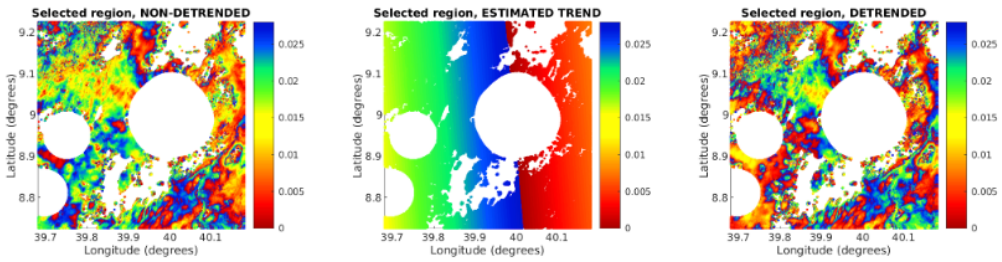
Step 7 – Prepare model bounds and input file

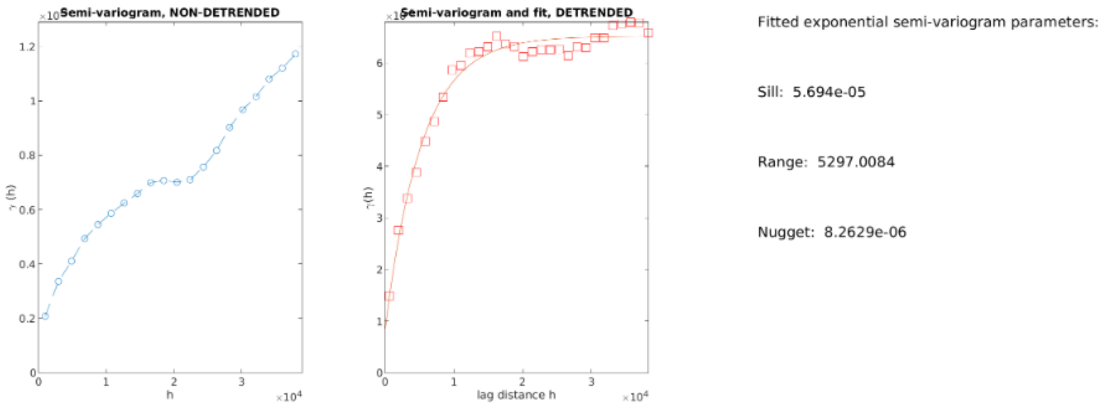
Outputs saved in the “InputFiles” folder:

- Volcano_Frame_RUNID.inp – Textfile to be used as input for GBIS in Step 8.

Outputs saved in the “Variograms” folder:

- Volcano_Frame_No_SS_RUNID.png – Experimental variogram fitted to the non-deforming (far-field) areas of the deformation image to characterise the noise characteristics of the image.

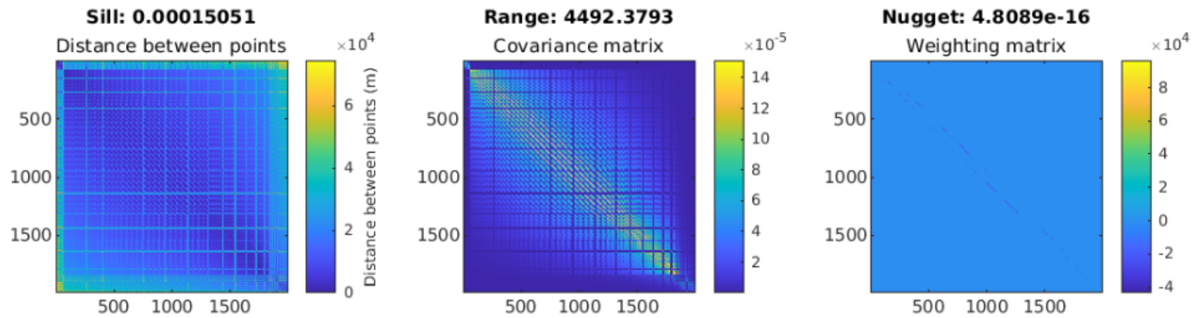


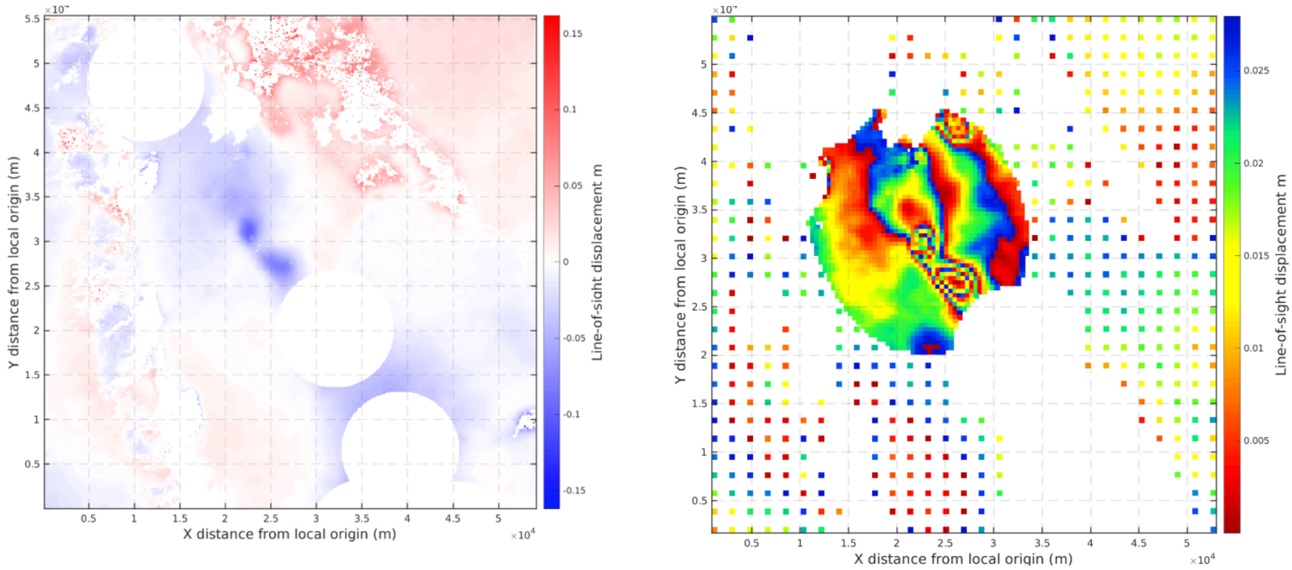


Step 8 – GBIS model setup

Outputs saved in the "Inversion_Results" folder:

- Invert_numFrames_ModelCode.mat (e.g. invert_1_M.mat) – GBIS Inversion results (see GBIS manual) – also contains details of InSAR input data and misfits.
- SeedingStats.mat – If seeding runs are used, this file shows the start, minimum and maximum model bounds for each parameter for each seeding run, and the associated reduction in the model bounds from the initial wide bounds.
- In "Figures" folder:
  - Covariance_Volcano_Frame_RUNID.png – Distance between points, covariance matrix and weighting matrix with noise characteristics
  - Unwrapped_Volcano_Frame_RUNID.png – Unwrapped input data (downsampled and full resolution)
  - Wrapped_Volcano_Frame_RUNID.png – Wrapped input data (downsampled and full resolution)

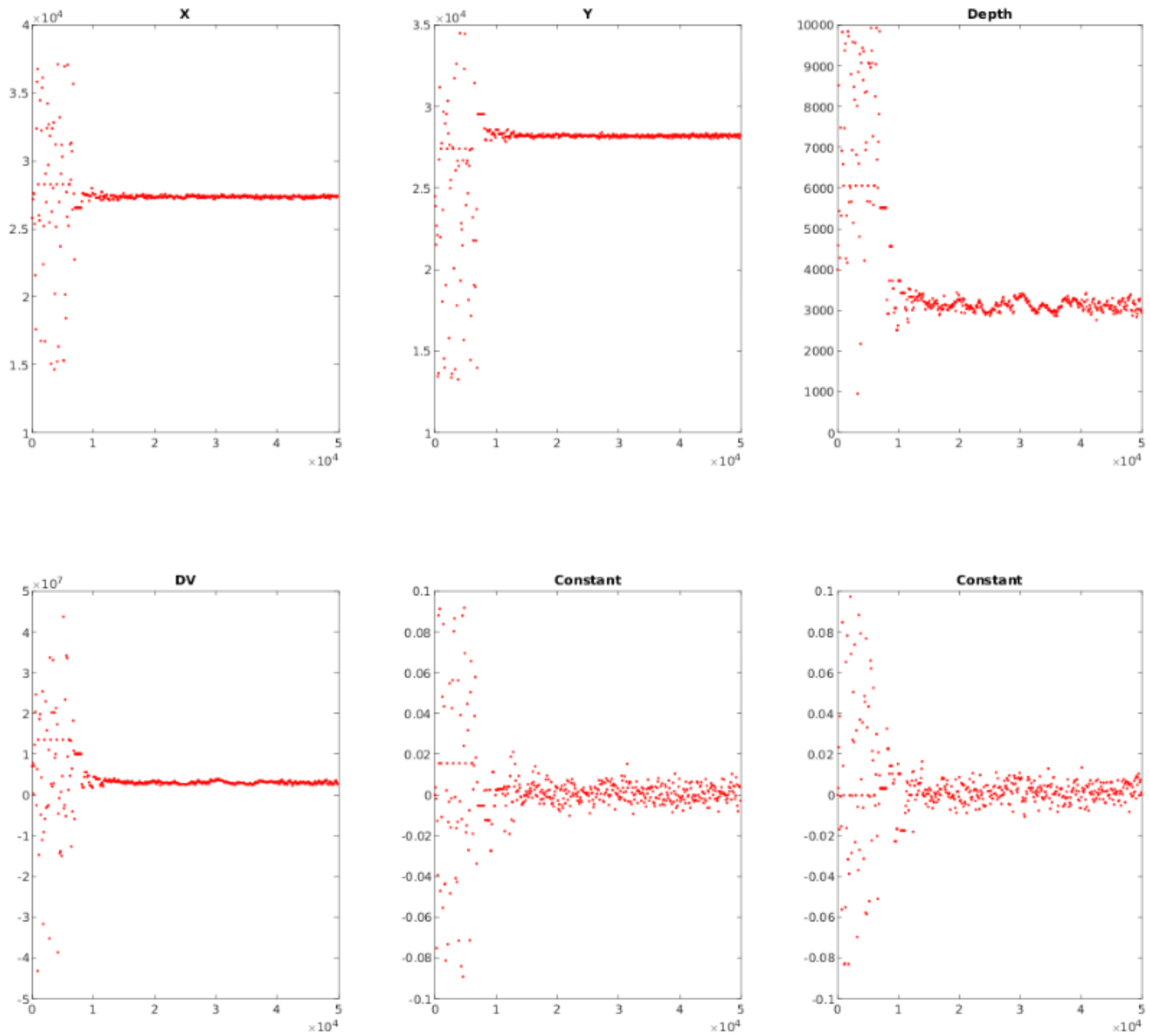




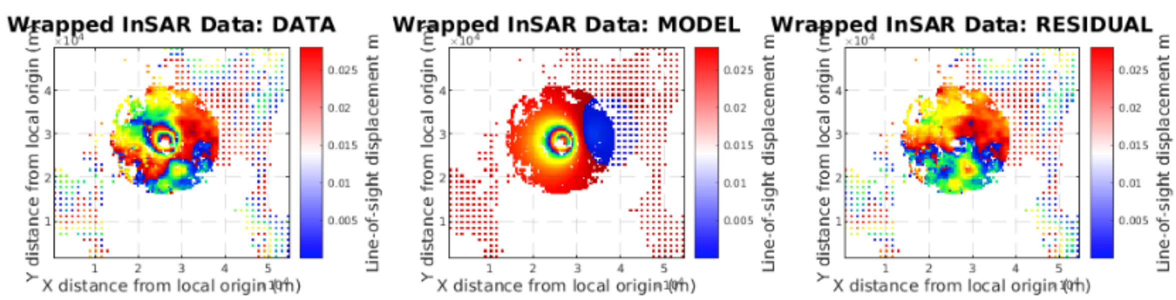
Step 9 – Create reports

Outputs saved in the “Summary_Figures” folder:

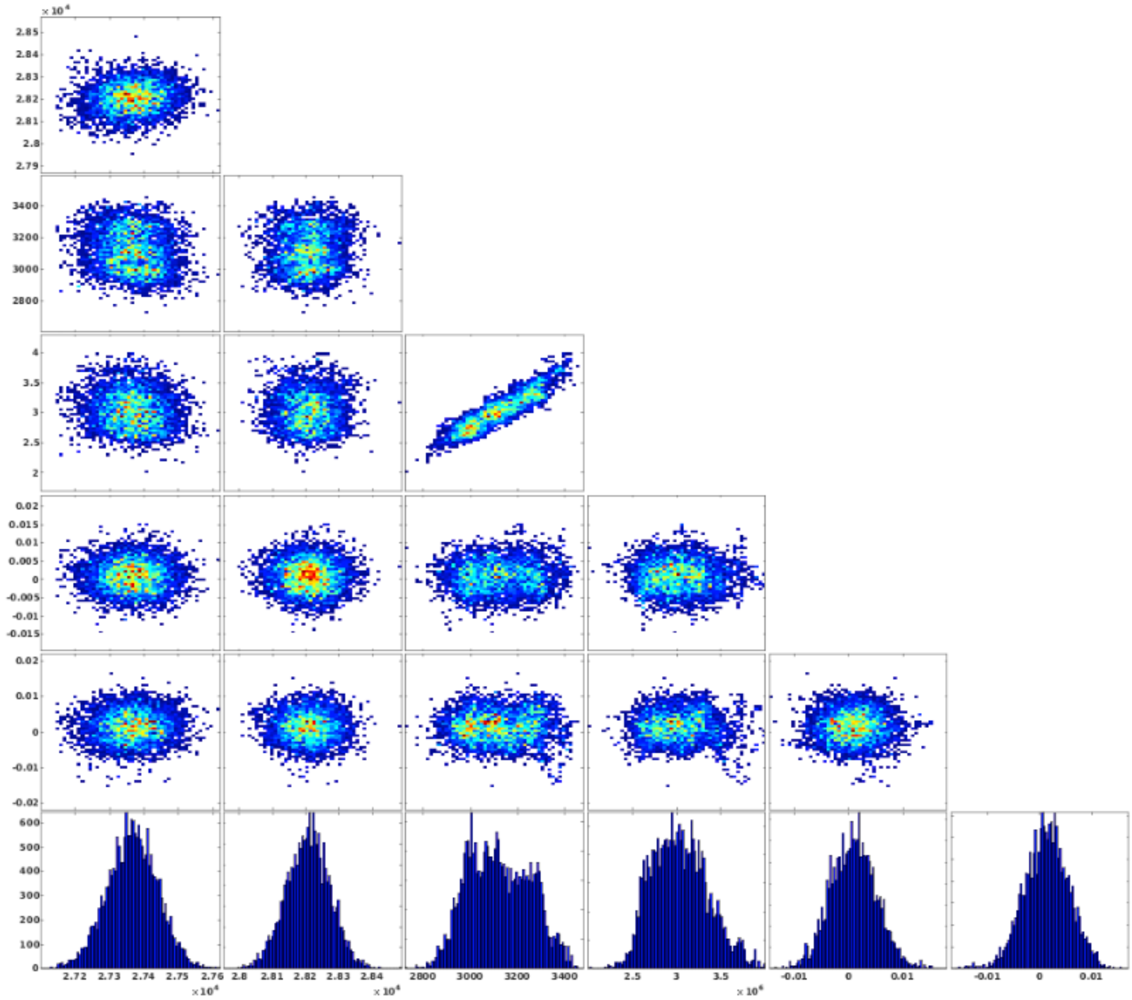
- Convergence.png – Convergence charts for all model parameters



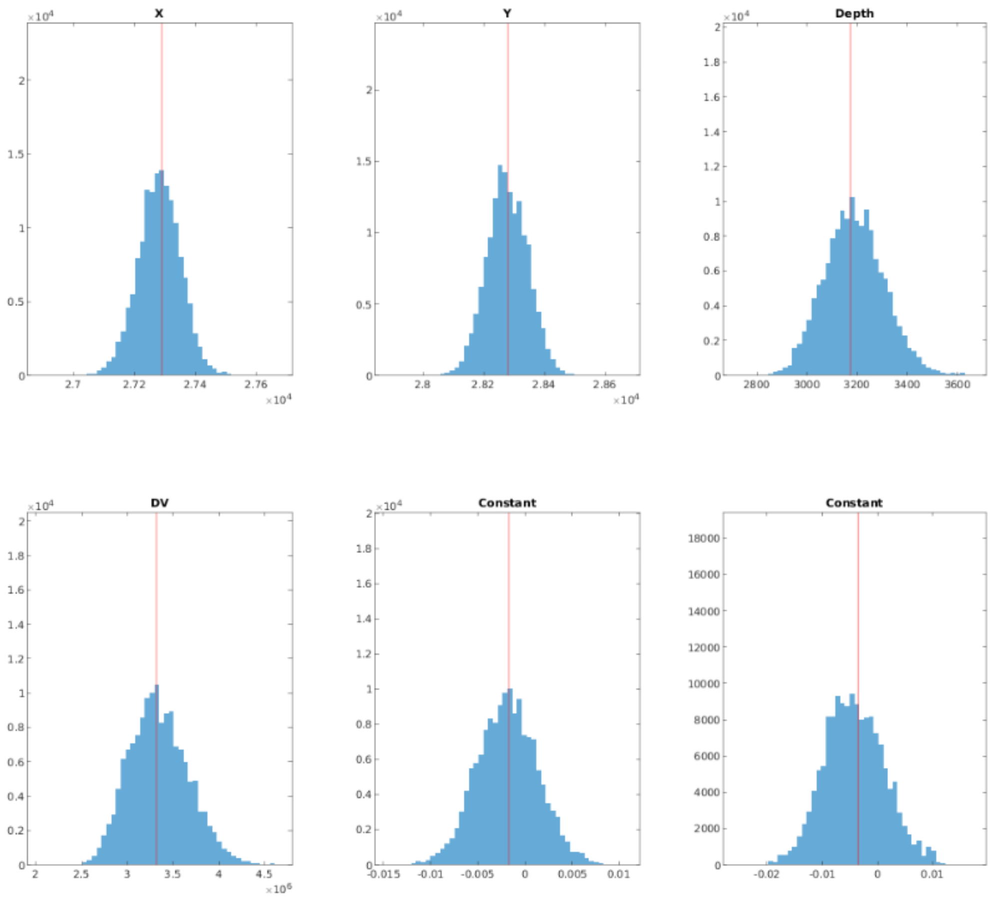
- InSAR_Data_Model_Residual_Volcano_Frame_RUNID.png – Data-model-residual plot for the best-fit model (Wrapped/unwrapped/downsampled/full-resolution)



- JointProbabilities.png – Trade-off plots between all model parameters



- PDFs.png – PDFs of model parameters tested during the inversions



Outputs saved in the “PDF_Reports” folder:

- Volcano_Frame_RUNID_ModelCode.pdf – PDF report containing all pre-processing and inversion results figures, with best-fit parameters recorded.

Step 10 – Create deformation catalogue

Outputs saved in the “Deformation_Catalogues” folder

- BULKRUNID_RUNID.csv – Deformation catalogue containing all temporal, spatial and source parameters from all runs in a bulk run (for all signals and all model geometries tested).

Deformation Sources

You can choose which source(s) to run and compare and their model bounds by editing the Step 7 section of the input options (see optional parameters section). The currently available sources in GBIS BULK and their model codes are listed below. For full details and parameters, consult the ‘optional parameters’ section and scripts in the ‘deformation sources’ sub-folder.

- Point source (Mogi, 1958) ‘M’

- Rectangular Dislocation Dyke or Sill (Okada, 1986) ‘D’ or ‘S’
- Spherical source (McTigue, 1987) ‘T’
- Penny-shaped crack (Fialko, 2001) ‘P’
- Prolate spheroid (Yang, 1988) ‘Y’
- Horizontal Crack (Sun, 1969) ‘C’ (pressure change) or ‘V’ (volume change)
- Spheroid (Cervelli, 2013) ‘E’
- Compound Dislocation Model, CDM (Nikkhoo, 2017):
    - Full geometric freedom - ‘A’
    - Spherical – ‘B’
    - Square sill – ‘I’
    - Elongated sill - ‘J’
    - Vertically planar (dyke) – ‘K’
    - Prolate geometry – ‘L’
    - Oblate geometry – ‘O’

Download

The latest available version is available on GitHub (https://github.com/ben-ireland/GBIS_V1.1_BULK) whilst stable versions are archived on Zenodo (doi: 10.5281/zenodo.18303435)

References

Cervelli, P.F., 2013, December. Analytical expressions for deformation from an arbitrarily oriented spheroid in a half-space. In *AGU Fall Meeting Abstracts* (Vol. 93, pp. 12-13).

Fialko, Y., Khazan, Y. and Simons, M., 2001. Deformation due to a pressurized horizontal circular crack in an elastic half-space, with applications to volcano geodesy. *Geophysical Journal International*, 146(1), pp.181-190.

McTigue, D.F., 1987. Elastic stress and deformation near a finite spherical magma body: resolution of the point source paradox. *Journal of Geophysical Research: Solid Earth*, *92*(B12), pp.12931-12940.

Mogi, K., 1958, Relations between the eruptions of various volcanoes and the deformations of the ground surfaces around them. *Bulletin of the Earthquake Research Institute*, vol. 36, 99-134.

Nikkhoo, M., Walter, T.R., Lundgren, P.R. and Prats-Iraola, P., 2016. Compound dislocation models (CDMs) for volcano deformation analyses. *Geophysical Journal International*, p.ggw427.

Sun, R.J., 1969. Theoretical size of hydraulically induced horizontal fractures and corresponding surface uplift in an idealized medium. *Journal of Geophysical Research*, *74*(25), pp.5995-6011.

Yang, X.M., Davis, P.M. and Dieterich, J.H., 1988. Deformation from inflation of a dipping finite prolate spheroid in an elastic half-space as a model for volcanic stressing. *Journal of Geophysical Research: Solid Earth*, *93*(B5), pp.4249-4257.